

CHAPTER 1

Introduction



Objectives:

- ❑ To define the Turing model of a computer.**
- ❑ To define the von Neumann model of a computer.**
- ❑ To describe the three components of a computer.**
- ❑ To list topics related to computer hardware.**
- ❑ To list topics related to data.**
- ❑ To list topics related to software.**
- ❑ To discuss some social and ethical issues.**
- ❑ To give a short history of computers.**

1.1 TURING MODEL

The idea of a universal computational device was first described by Alan Turing in 1937. He proposed that all computation could be performed by a special kind of a machine, now called a Turing machine. Although Turing presented a mathematical description of such a machine, he was more interested in the philosophical definition of computation than in building the actual machine.

Figure 1.1: A single purpose computing machine

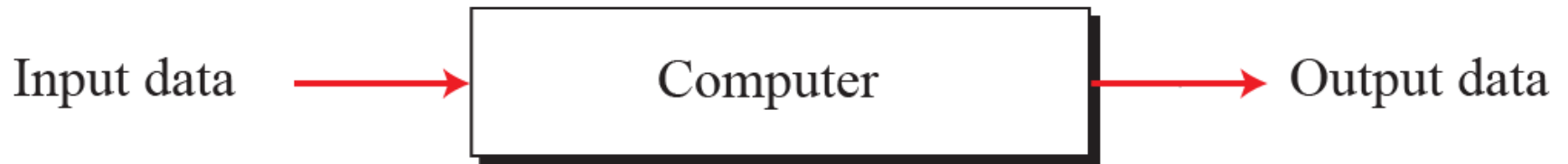


Figure 1.2: A computer based on the Turing model:

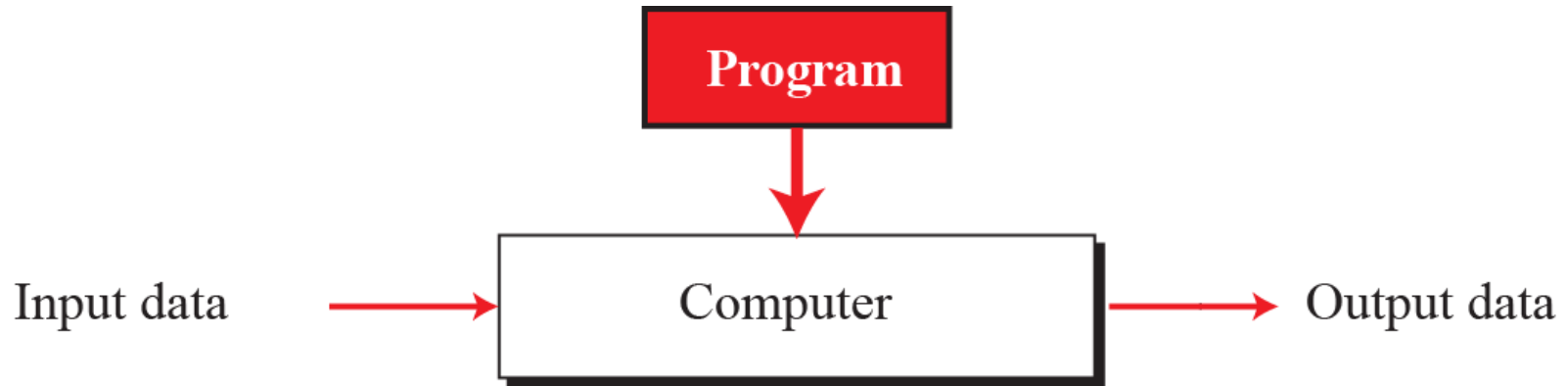


Figure 1.3: The same program, different data

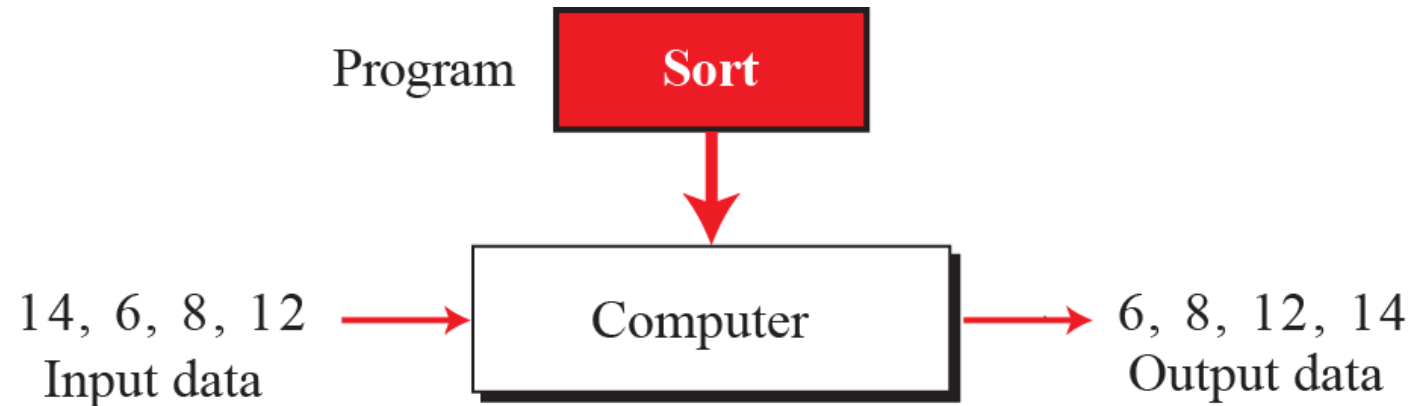
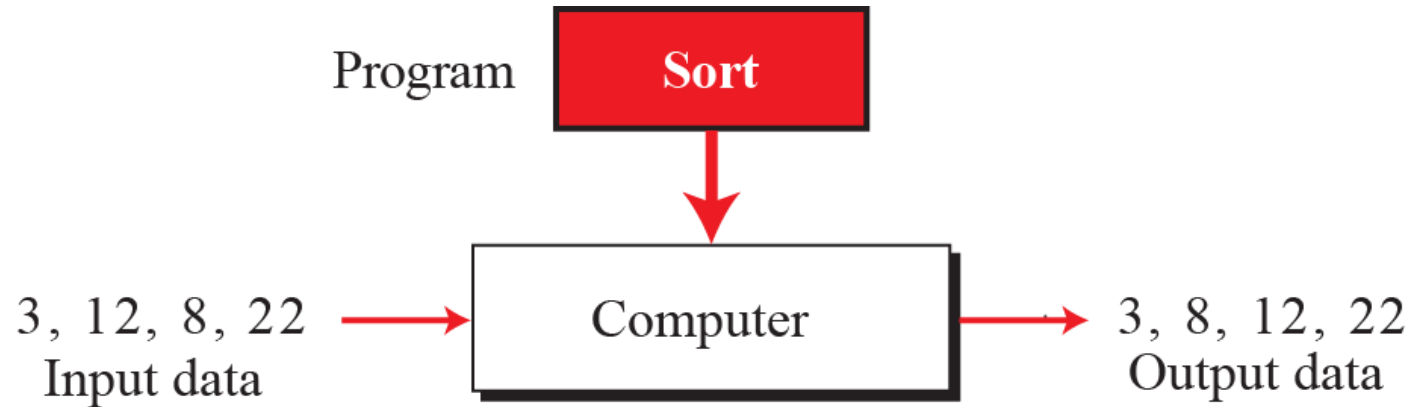
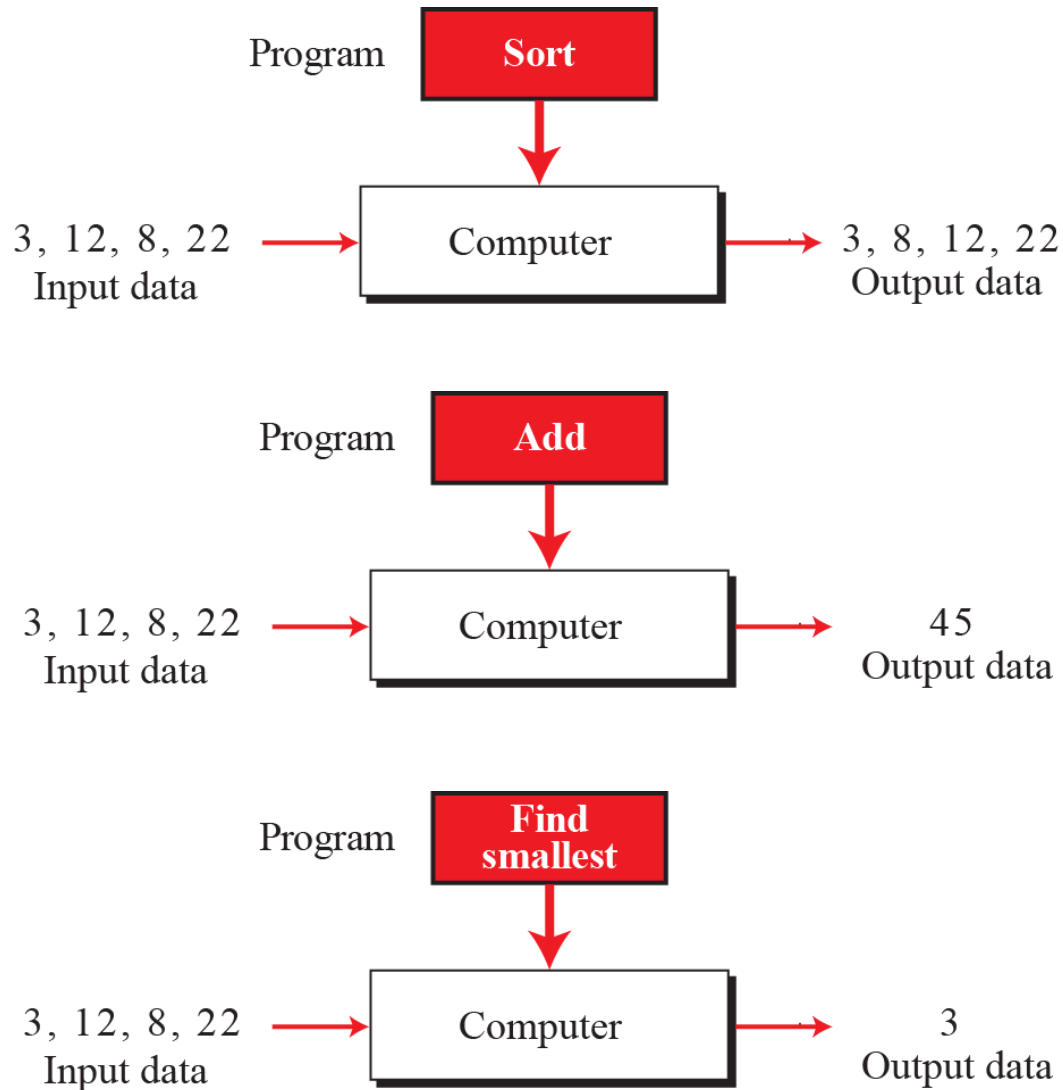


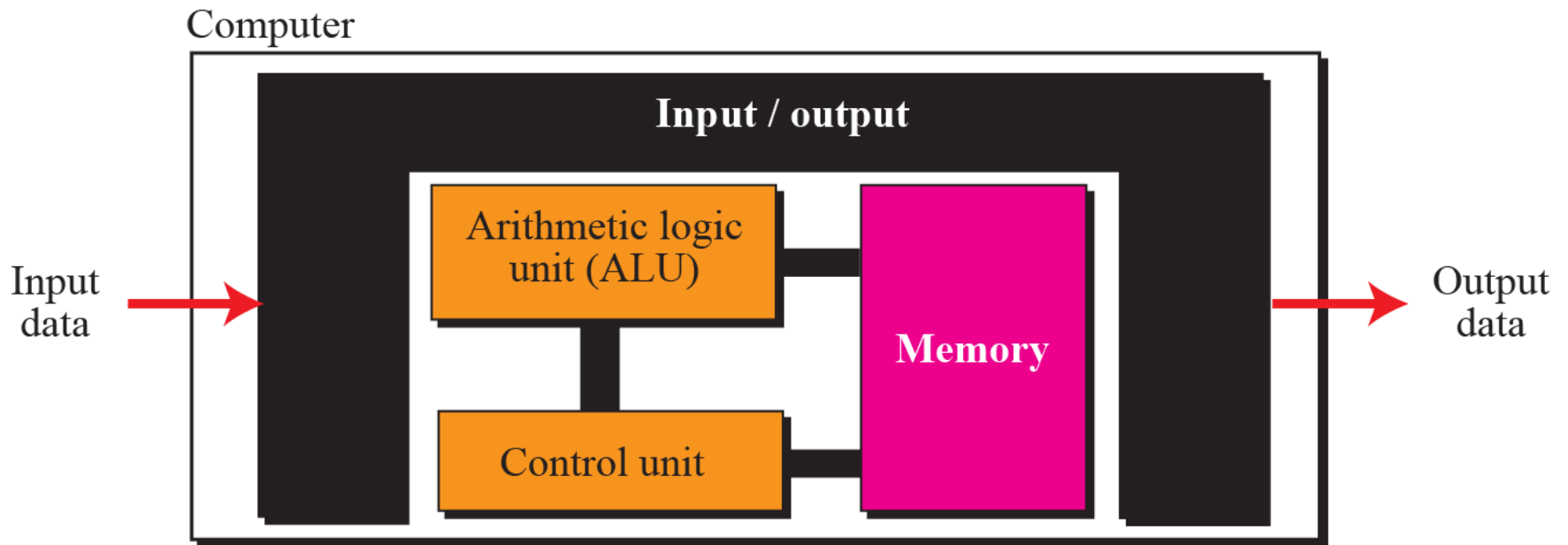
Figure 1.4: The same data, different programs



1.2 VON NEUMANN MODEL

Computers built on the Turing universal machine store data in their memory. Around 1944–1945, John von Neumann proposed that, since program and data are logically the same, programs should also be stored in the memory of a computer.

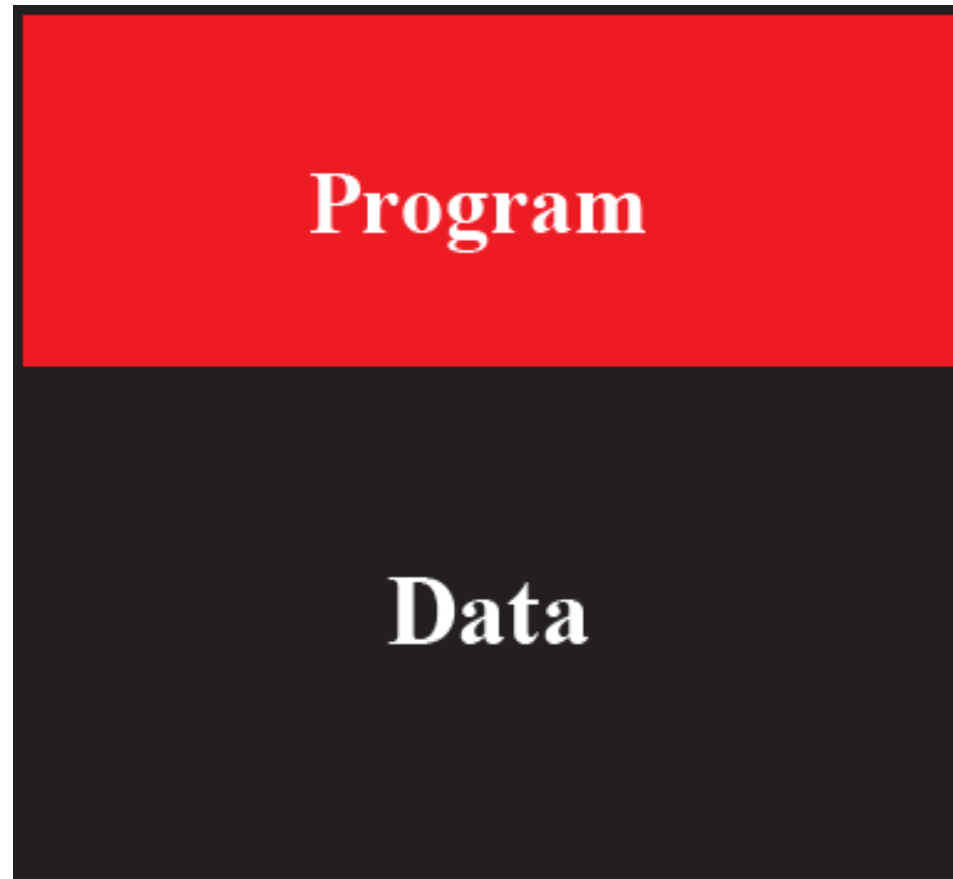
Figure 1.5: The Von Neuman model



1.3 COMPUTER COMPONENTS

We can think of a computer as being made up of three components: computer hardware, data, and computer software.

Figure 1.6: Program and data in memory



Memory

Figure 1.7: A program made of instructions

- 1. Input the first number into memory.**
- 2. Input the second number into memory.**
- 3. Add the two together and store the result in memory.**
- 4. Output the result.**

Program

1.4 HISTORY

In this section we briefly review the history of computing and computers. We divide this history into three periods.

Mechanical machines (before 1930)

The birth of electronic computers (1930–1950)

Computer generations (1950–present)

1.5 SOCIAL AND ETHICAL ISSUES

Computer science has created some peripheral issues, the most prevalent of which can be categorized as social and ethical issues.

1.6 Computer Science As Discipline

With the invention of computers, a new discipline has evolved: computer science. Like any other discipline, computer science has now divided into several areas. We can divide these areas into two broad categories: systems areas and applications areas.

computer science trends

Artificial Intelligence / Machine Learning/Deep Learning

Big Data

Bioinformatics / Medical Technology

Cloud Computing

Computer Vision / Natural Language Processing

Cybersecurity

computer science trends

Internet of Things

Quantum Computing

Robotics

Virtual Reality

<https://www.create-learn.us/blog/top-computer-science-trends/>

<https://www.computer.org/publications/tech-news/trends/2022-report>